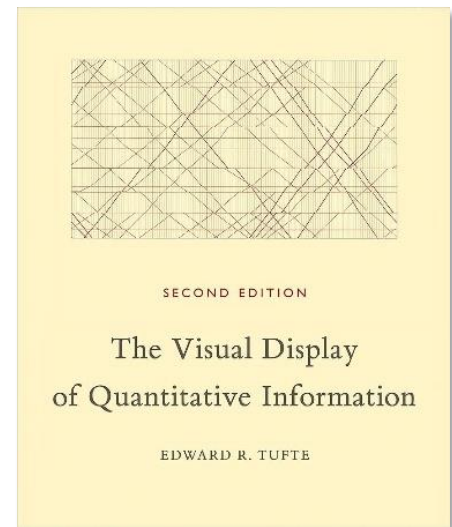


# Lecture 4. Visualising Data

## What makes a good graph?

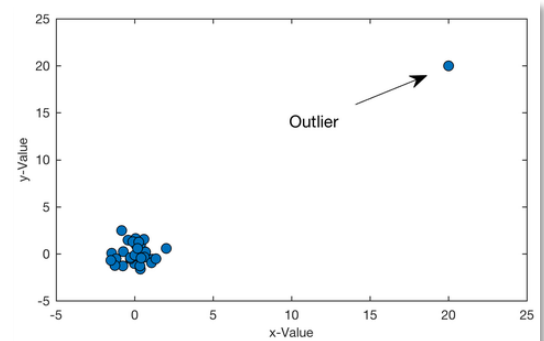
1. As simple and easy to interpret as possible
2. Make important information stand out
3. Clear axis labels, legend keys and title or caption
4. Show underlying data points/distributions (when possible)
5. Don't distort differences by manipulating the scale



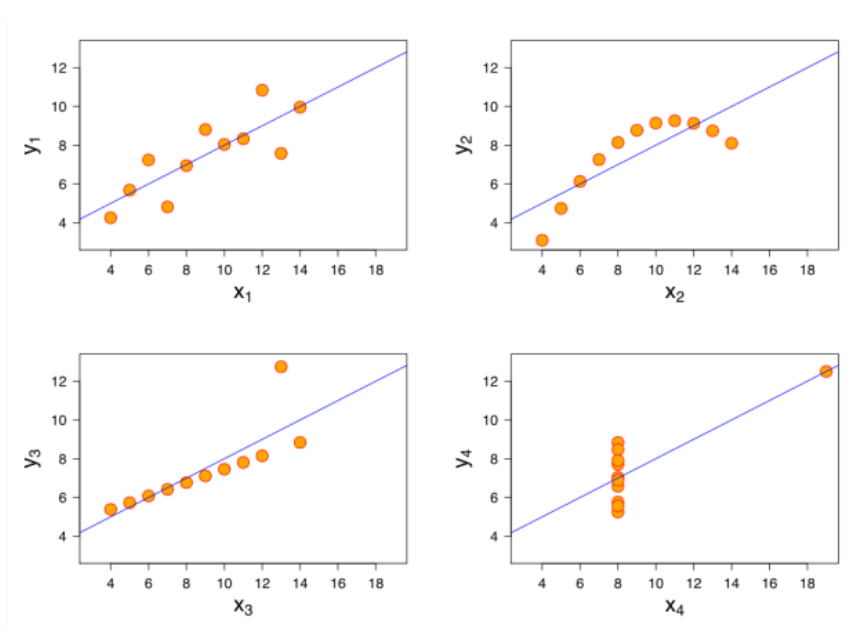
## Visualise early!

You should always visualise your data **before** you conduct data analysis

- Look for **outliers** (extreme scores)
- Look for data entry **mistakes**
- Check the shape of **distribution**
- Visualise **relationships** between variables



**Example:** Anscombe's quartet. These four sets of variables have exactly the same numerical **strength** of association ( $r=.82$ ), but when we visualise them, we discover that the four associations are very different in nature:



## Data visualisation in R

We are going to be using the package **ggplot2** which is part of **tidyverse** for creating graphs in R

You can explore different graphs with code to create them on the r graph gallery website: <https://www.r-graph-gallery.com/ggplot2-package.html>



### What type of graph?

First work out what you are trying to show:

- Single variable or the relationship between variables?
- Are the variables numeric or categorical?

The website “Data to Viz” is a great resource for exploring the different graph options depending on your data. <https://www.data-to-viz.com/>

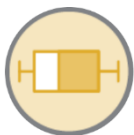
### Single numeric variable:



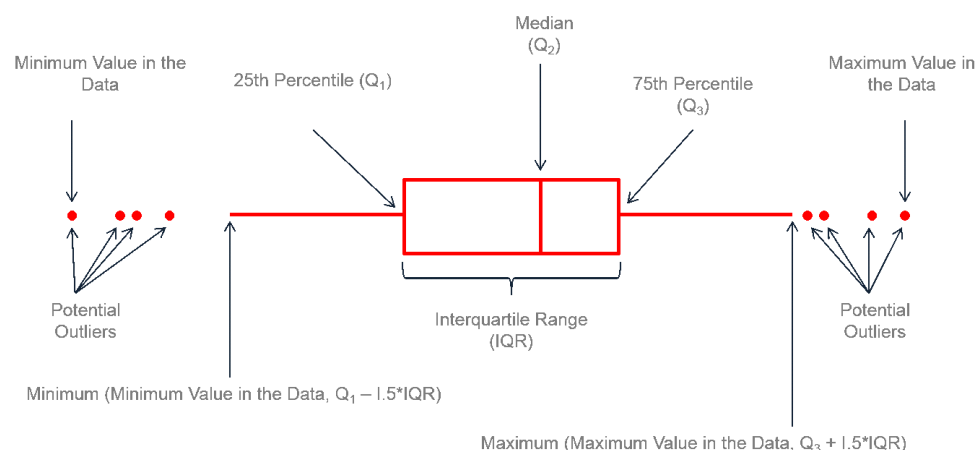
**Histogram:** Often used for checking the distribution of data, e.g., whether the data follow a “normal distribution”.



**Density plot:** A “smoothed” version of the histogram. Often used when we are interested in the overall “shape” of the data, rather than precise counts for individual values.



**Box plot:**



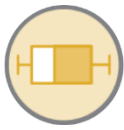
### Single categorical variable



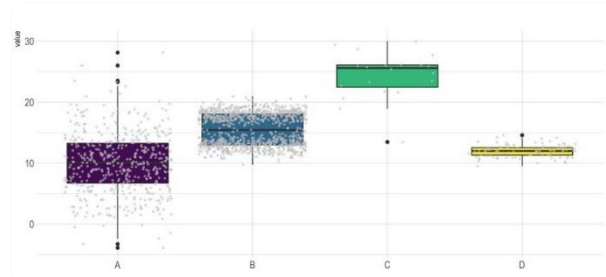
**Bar chart:** used to visualise a single categorical variable – frequencies, percentages or proportions of items in categories.

We don’t use pie charts – we’re not good at judging angles, so pie charts are usually not recommended as a way of representing count data.

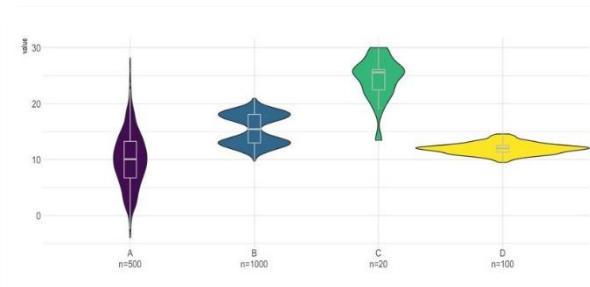
## One numeric and one categorical variable



**Box plot:** visualises M and SD of the continuous variable. However, it can obscure some important details about the data distribution, especially if it is not normally distributed.

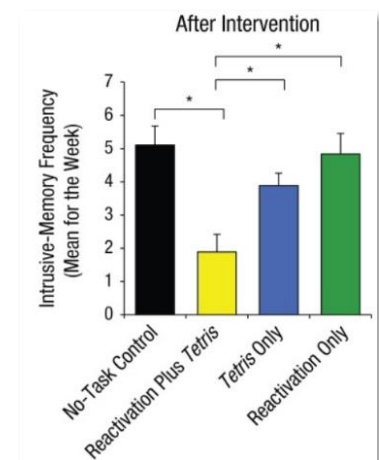


**Violin plot:** capable of depicting the underlying distribution of the data.



Why not bar charts?

Bar charts are often used in psychology papers to show the mean of a numeric variable for each level of one or more categorical variables. However, they can hide important details about the underlying distribution of data. Therefore, we recommend using a combination of boxplot and violin plot rather than a bar chart.



## Two numeric variables



**Scatterplot:** Useful for visualising a relationship between two continuous variables. Each point represents an observation. The black line and grey band that go through the points are optional (these are the line of best fit and the confidence intervals – more on this in the lecture on linear regression).

